Stochastic Calculus

See
- Ch 4 of the lecture note
- Øksendal "Stochastic differential equation"

Motivation : How to represent random processes in continuous time ?

It is often said that white noise is the time derivative of the Brownian motion W_t.

So its tempting to write

$$\frac{dX_t}{dt} = AX_t + \text{``}\frac{dW_t}{dt}\text{''} . \quad \text{— ①}$$

But Brownian motion is not differentiable !

So let us write ① in an integral form

$$X_t = X_0 + \int_0^t AX_s\,ds + \text{``}\int_0^t \frac{dW_s}{ds}\,ds\text{''} \quad \text{— ②}$$

② still contains an "illegal" expression $\frac{dW_s}{ds}$.

But we can define a new concept $\int_0^t dW_s$ (called stochastic integral) to write ② as

③

We also decide to write ③ more compactly as

④

④ is called the stochastic differential equation (SDE).

More generally, SDE can be of the form

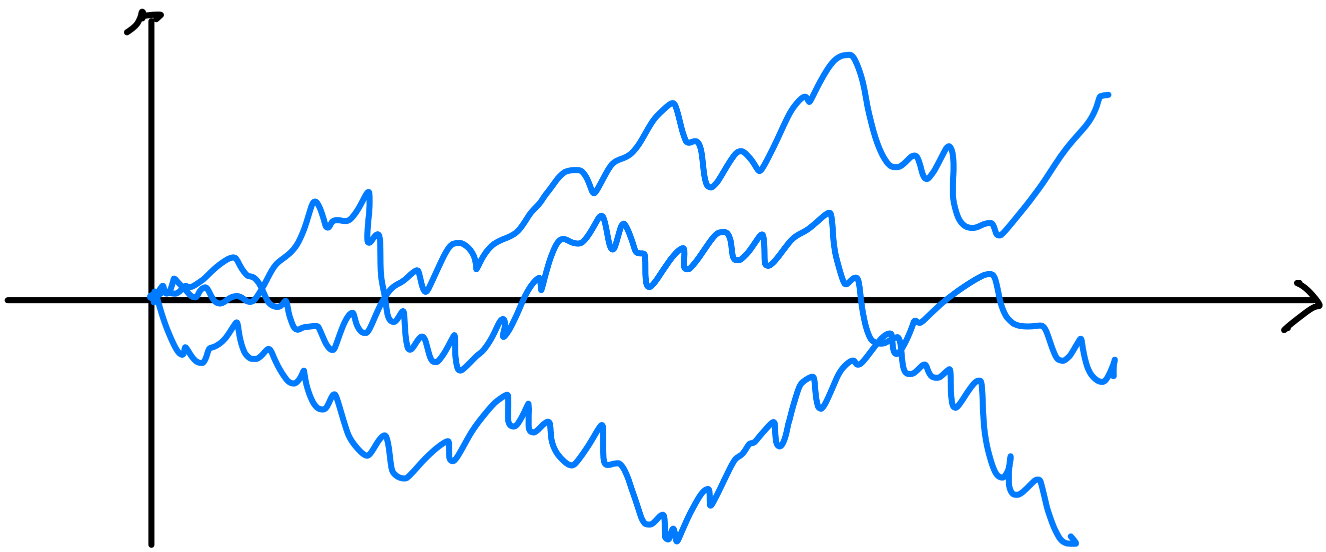
which is a short-hand representation of

Q: How to define stochastic integrals?

A1 : Ito integral

A2 : Stratonovich integral

Standard Brownian motion W_t



Properties:

(1) W_t is continuous w.p. 1

(2) W_{t_1}, $W_{t_2} - W_{t_1}$, $W_{t_3} - W_{t_2}$, ...
are independent

(3) $W_t - W_s$ is zero-mean Gaussian

Q: How to simulate W_t?

Let Δt be the step size. Then

$$W_{t+\Delta t} = W_t + \sqrt{\Delta t}\, V_t$$

$$V_t \overset{iid}{\sim} N(0, 1).$$

Ito integral

Let W_t be the standard Brownian motion.
We define the Ito integral by

The RHS converges in the mean-square sense if g is sufficiently regular.

Property (Ito isometry)

$$\mathbb{E}\left[\left(\int_0^t g_s \, dW_s\right)^2\right] = \mathbb{E}\left[\int_0^t g_s^2 \, dt\right]$$

Proof (informal)

$$(\text{LHS}) \approx \mathbb{E}\left[\sum_{i=1}^{n} g_{t_i} (W_{t_{i+1}} - W_{t_i})^2\right]$$

(2)

$$= \mathbb{E} \sum_{i=1}^{n} g_{t_i}^2 (W_{t_{i+1}} - W_{t_i})^2$$

(3)

$$= \mathbb{E} \sum_{i=1}^{n} g_{t_i}^2 (t_{i+1} - t_i)$$

$$\approx \mathbb{E} \int_0^t g_s^2 \, ds$$

The process defined by

$$X_t = x_0 + \int_0^t f_s \, ds + \int_0^t g_s \, dW_t$$

or equivalently

$$dX_t = f_t \, dt + g_t \, dW_t$$

is called the Ito process.

The Ito process can be numerically simulated by the Euler-Maruyama method:

Q: Suppose we have an Ito SDE

$$dX_t = f_t \, dt + g_t \, dW_t .$$

Suppose $Y_t = h(t, X_t)$ is a function of X_t.

What SDE does Y_t satisfy?

$$dY_t = \boxed{\quad ? \quad} \, dt + \boxed{\quad ? \quad} \, dW_t$$

Ito formula

If X_t is the Ito process and
$Y_t = h(t, X_t)$, then

$$dY_t = \frac{\partial h}{\partial t} dt + \frac{\partial h}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 h}{\partial x^2} (dX_t)^2$$

where $(dX_t)^2$ is computed based on the rules:

$$(dt)^2 = 0, \quad (dt)(dW_t) = 0, \quad (dW_t)^2 = dt.$$

Proof: Follows from the Ito isometry.

Example Suppose
$$dX_t = \alpha(t, X_t)\, dt + \beta(t, X_t)\, dW_t.$$
and $Y_t = h(X_t) = X_t^2$.

$\Rightarrow$

So, the SDE for Y_t is

— ⑤

As a special case, suppose

$$\alpha(t, X_t) = aX_t, \quad \beta(t, X_t) = 1.$$

and let $\Sigma_t := \mathbb{E} X_t^2$.

Taking $\mathbb{E}$ of ⑤:

$$\mathbb{E}[dY_t] = \mathbb{E}[(2X_t\alpha + \beta^2)]\,dt$$

$$\mathbb{E}[dX_t^2] = \mathbb{E}[(2aX_t^2 + 1)]\,dt$$

$$\frac{d}{dt}\mathbb{E}[X_t^2] = \mathbb{E}[(2aX_t^2 + 1)]$$

$$\frac{d}{dt}\Sigma_t = 2a\Sigma_t + 1$$ (Lyapunov equation)

Continuous-Time SOC

- See Ch 5 of the lecture note

Assume control-affine dynamics

$$dX_t = [f_t(X_t) + g_t(X_t)\, u_t]\, dt + \sigma_t(X_t)\, dW_t$$

$$X_0 = x_0$$

Problem:

$$\min_{u[0,T]} \mathbb{E}\left[\int_0^T \left(V_s(X_s) + \frac{1}{2} u_s^T R_s u_s \right) dt + \psi(X_T) \right]$$

Stochastic HJB equation

Define the cost-to-go function

$$C(x, t, u_{[t,T]}) = \mathbb{E}\left[\int_t^T \left(V_s(X_s) + \frac{1}{2} u_s^T R_s u_s \right) ds + \psi(X_T) \right]$$

and the value function

$$J(x,t) = \min_{u[t,T]} C(x, t, u_{[t,T]})$$

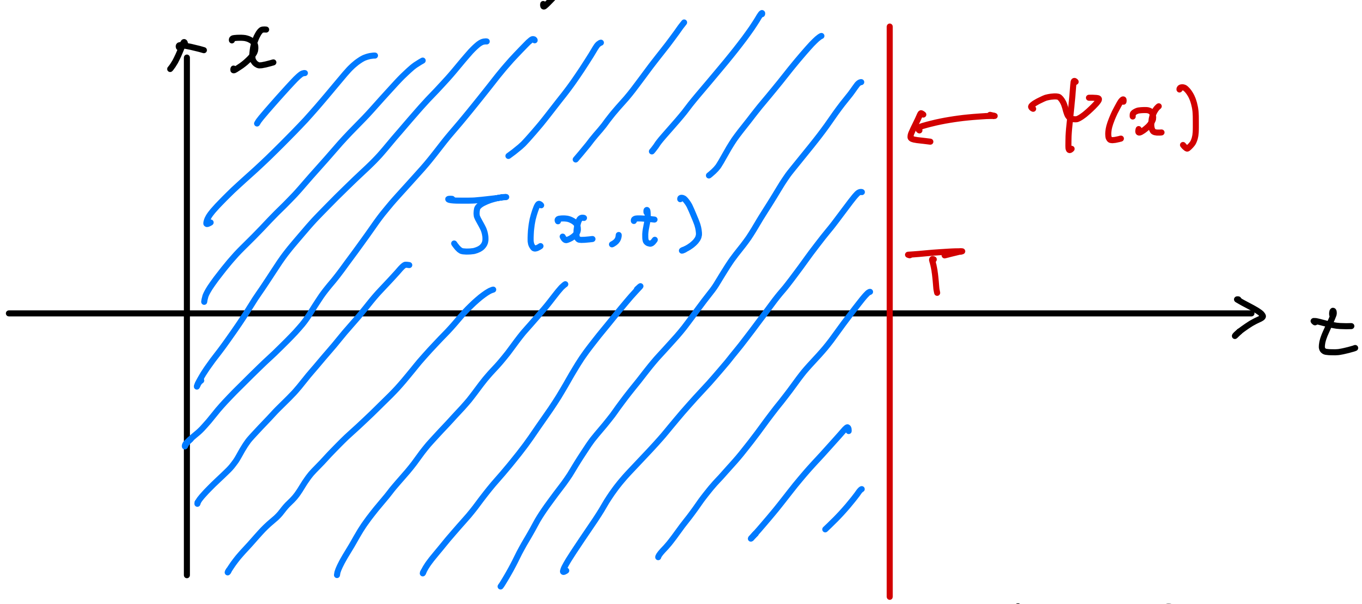
Theorem 5.1 (Verification Theorem)

Suppose

(a) $\partial_t J = \frac{\partial J}{\partial t}$, $\partial_x J = \frac{\partial J}{\partial x}$, $\partial_x^2 J = \frac{\partial^2 J}{\partial x^2}$ exist

(b) J satisfies the PDE

with the boundary condition $J(x,T) = \psi(x)$.



Then (i) $J(x,t)$ is the value function

(ii) Optimal control policy is

$$u_t^* = -R_t^{-1} g_t^T \partial_x J.$$

Proof Application of Ito formula.